

§4-4 Indeterminate Forms and L'Hospital's Rule

Homework : 5,9,11,15,17,26,47,49,50,51,59,73,74

(I) Indeterminate forms: (不確定型)

$$\frac{0}{0}, \frac{\infty}{\infty} = \infty \cdot (0), 1^\infty, 0^0, \infty^0, \infty - \infty$$

Determinate forms

$$\infty + \infty = \infty, (\infty)(\infty) = \infty, \frac{3}{5}$$

(II) L'Hospital's Rule

(i) $\frac{f}{g}$: indeterminate form

(ii) $\frac{f'}{g'}$: determinate form

$$\Rightarrow \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

解題常遇的狀況.

(iii) $\frac{f'}{g'}$: indeterminate $\Rightarrow$ same procedure can be continued. (見 Examples 2,6,8)

(iv) $1^\infty, 0^0, \infty^0 \Rightarrow$ Take $\ln \Rightarrow \frac{f}{g}$ (見 Examples 4,5)

Example 1 : $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{2x}{1} = 2.$$

Example 2 : $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1} = \lim_{x \rightarrow 1} \frac{1}{x} = 1.$

Example 3 : $\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \lim_{x \rightarrow \infty} \frac{e^x}{2x} = \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty.$

Generally speaking :

$x^x \ll x! \ll e^x \ll x^n \ll \ln x$ when x large

Example 4 : $\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x}$.

Solution : $y = (1 + \sin 4x)^{\cot x}$

$$\ln y = \cot x \ln(1 + \sin 4x)$$

$$= \frac{\ln(1 + \sin 4x)}{\tan x}$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \ln y &= \lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 4x)}{\tan x} \\ &= \lim_{x \rightarrow 0^+} \frac{4 \cos 4x}{(\sec^2 x)(1 + \sin 4x)} = 4 \end{aligned}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} y = e^4.$$

Example 5 : $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$.

Solution : $y = \left(1 + \frac{1}{x}\right)^x$

$$\ln y = x \ln \left(1 + \frac{1}{x}\right) = \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Example 6 : $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2}$

$$= \lim_{x \rightarrow 0} \frac{2 \sec x \tan x}{6x} = \frac{1}{3}.$$

Example 7 : If f' is continuous, $f(2) = 0$, $f'(2) = 7$, evaluate

$$\lim_{x \rightarrow 0} \frac{f(2+3x) + f(2+5x)}{x}.$$

Solution :

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(2+3x) + f(2+5x)}{x} \\ &= \lim_{x \rightarrow 0} \frac{3f'(2+3x) + 5f'(2+5x)}{1} \\ &= 3f'(2) + 5f'(2) = 56. \end{aligned}$$

Example 8 : For what values of a and b is the following equation true?

$$\lim_{x \rightarrow 0} \left(\frac{\sin 2x}{x^3} + a + \frac{b}{x^2} \right) = 0.$$

Solution :

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 2x + ax^3 + bx}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{2\cos 2x + 3ax^2 + b}{3x^2} \quad \left(= \frac{f}{g} \right) \end{aligned}$$

$\frac{f}{g}$ must be indeterminate form

$$\Rightarrow 2 + b = 0 \Rightarrow b = -2$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{-4\sin 2x + 6ax}{6x} \\ &= \lim_{x \rightarrow 0} \frac{(8-6a)x}{6x} = \lim_{x \rightarrow 0} \frac{8-6a}{6} = 0 \\ &\Rightarrow a = \frac{4}{3}. \end{aligned}$$